Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	iPSC-derived Ocular Regenerative Therapies and Mitochondrial Enhancement Platforms			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Cell and Tissue Therapeutics	Bio-processes / Bio-Equipment	Bio-process Technologies
Concept of Proposed Technology	Ocular diseases involving irreversible loss or dysfunction of specialized cell populations remain major causes of visual impairment worldwide. Donor tissue shortages, immune rejection, and variable long-term outcomes continue to limit existing transplantation-based approaches. This joint R&D program seeks proposals focused on scalable, standardized, and functionally enhanced iPSC-derived ocular cell therapies that: • Provide renewable and consistent sources of clinically relevant ocular cells; • Demonstrate robust functional performance and stress resilience after transplantation; • Are compatible with clinically feasible delivery methods and GMP-aligned manufacturing; • Support both therapeutic transplantation and human-cell-based disease modeling or drug screening. In addition, applicants are encouraged to explore mitochondrial or cellular bioenergetics enhancement strategies including, but not limited to, mitochondrial conditioning, metabolic modulation, or protective signaling activation to improve durability and therapeutic consistency of transplanted cells.			

Rationale for Joint R&D Project	Development of clinically viable ocular cell therapies requires coordinated progress in cell differentiation, purification, functional validation, preclinical modeling, and manufacturing process design. • Korean startups/SMEs contribute strengths in scalable cell production, formulation, quality control, regulatory preparation, and commercialization. • U.S. research institutions provide advanced expertise in ocular disease biology, functional phenotyping, in vivo and microphysiological models, and translational research methodologies. A Korea–US joint R&D framework enables efficient maturation of iPSC-based ocular therapies from laboratory feasibility to preclinical and regulatory-ready platforms, while establishing globally competitive development pipelines.
Key Words	1. Induced Pluripotent Stem Cells (iPSC) 2. Corneal Endothelial Cells 3. Cell-Based Therapeutics 4. Mitochondria-Derived Peptides (Humanin, MOTS-c)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3-4)	**End of the Project**	Level (5)
R&D Objectives	Proposals should aim to: 1. Establish robust differentiation and purification processes for iPSC-derived ocular cell types; 2. Demonstrate functional potency and stress resilience relevant to therapeutic mechanisms; 3. Validate safety and preliminary efficacy in appropriate preclinical models; 4. Develop manufacturing-aligned processes and quality frameworks; 5. Enable human-cell platforms for disease modeling or compound screening, where applicable. Projects are expected to progress from laboratory-scale feasibility (TRL 3–4) toward preclinical validation and process readiness (TRL 5).			
R&D Target Specifications	1. Standardized iPSC-derived Ocular Cell Generation • Demonstrate reproducible differentiation, purification, and identity of target ocular cell populations suitable for therapeutic use. 2. Functional Potency and Stress Resilience • Establish quantitative potency assays reflecting disease-relevant functions and demonstrate enhanced survival or performance under stress conditions. 3. Safety and Genetic Stability • Verify low residual pluripotency, genomic stability, and absence of tumorigenic risk in preclinical settings. 4. Preclinical Proof-of-Concept • Demonstrate functional benefit and tolerability in relevant animal or advanced in-vitro disease models. 5. Manufacturing and Regulatory Readiness • Develop GMP-aligned processes, quality attributes, and technical documentation supporting future regulatory submissions. 6. Human-Cell Platform Utility (Optional but Encouraged) • Establish applicability of derived ocular cells for compound screening or therapy optimization in a scalable assay format.			

R&D Plan	**YEAR 1 — Platform Establishment & Feasibility Validation (TRL 3–4)** **Goal:** Establish technical feasibility of iPSC-derived ocular cell production and functional enhancement strategies. • Establish **robust differentiation and purification workflows** for selected ocular cell types. • Define **quantitative identity and functional potency assays** relevant to therapeutic mechanisms. • Evaluate feasibility of **cell enhancement strategies** (e.g., metabolic or mitochondrial conditioning) to improve stress resilience and functional stability. • Perform early-stage **in-vitro functional and stress-response validation**. • Initiate development of **human-cell-based assay platforms** for disease modeling or therapy optimization (optional but encouraged). **Deliverables (indicative):** • Reproducible cell production protocols. • Preliminary potency and safety profiles. • Feasibility evidence for functional enhancement strategies. **YEAR 2 — Optimization, Process Alignment & Preclinical Proof-of-Concept (TRL 4–5)** **Goal:** Finalize candidate cell processes and enhancement conditions that satisfy both therapeutic efficacy and manufacturing feasibility. • Optimize differentiation, purification, and enhancement conditions for **functional consistency and durability**. • Establish **manufacturing-aligned processes** and preliminary quality attributes. • Demonstrate **preclinical proof-of-concept** in relevant animal or advanced in-vitro disease models. • Validate **potency–efficacy correlations** between in-vitro assays and in-vivo or functional outcomes. • Expand and standardize **human-cell platforms** for compound screening or therapy optimization. **Deliverables (indicative):** • Optimized and scalable cell production process. • Preclinical efficacy and safety signals. • Defined critical quality attributes (CQAs). **YEAR 3 — Robustness, Regulatory Readiness & Translation Preparation (TRL 5)** **Goal:** Establish reliability, reproducibility, and regulatory readiness for clinical translation. • Demonstrate **multi-batch reproducibility and durability** of therapeutic performance. • Complete **biocompatibility, safety, and stability assessments** aligned with regulatory expectations. • Finalize **GMP-aligned manufacturing workflows and QC frameworks**. • Prepare **regulatory-supporting technical documentation** and development strategy. • Establish **commercialization and global partnership pathways**. **Deliverables (indicative):** • Regulatory-ready preclinical data package. • Manufacturing and quality systems framework. • Commercialization and clinical translation roadmap.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**CEC Biology & Translational Therapeutics Lead** - Co-Leads iPSC-CEC differentiation/maturation, characterization, and translational disease-relevance strategy - Co-leads potency assay definitions and safety rationale for ocular disease **Clinical/Surgical Translation & Evaluation Lead** - Defines clinical target product profile (TPP), clinically meaningful endpoints, surgical delivery considerations - Guides clinical-grade evaluation strategy for iPSC-CEC (and, as applicable, neuro-ophthalmic relevance) **Organ-on-a-Chip / MPS & Mechanobiology Lead** - Builds PSC-utilized ocular cell platform (microphysiological systems) for cornea/retina/optic nerve-relevant models - Enables high-throughput functional validation and mechanistic readouts under controlled microenvironments - Supports platform optimization using mechanobiology/biomaterials/nanotechnology interfaces
Expected R&R of Korean SME Partner	**CEC Biology & Translational Therapeutics Lead** - Co-Leads iPSC-CEC differentiation/maturation, characterization, and translational disease-relevance strategy - Co-leads potency assay definitions and safety rationale for ocular disease **Regulation** - Coordinates overall governance, milestones, and commercialization alignment - Optimize domestic manufacturing processes and apply GMP standards based on the introduced cell therapy technology - Conduct preclinical studies and prepare IND submissions in accordance with Korean regulatory requirements - Perform sterility testing, quality analyses, and other verification activities - Coordinate with clinical trial institutions and lead the application process for Advanced Regenerative Medicine clinical research - Develop commercialization strategies and enhance manufacturing capacity for future domestic and global markets

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Establish standardized differentiation, purification, and quantitative quality/potency frameworks for iPSC-derived corneal endothelial cells (iPSC-CECs). - Generate preclinical proof-of-concept showing iPSC-CECs restore corneal clarity/thickness and pump–barrier function under disease-relevant stress. - Demonstrate mitochondria/MDP-enhanced high-function ocular cells with improved bioenergetics and stress resilience (ΔΨm, ROS control, AMPK/TRIM72 signaling), translating into stronger CEC junction/pump performance and improved CEC robustness. - Build a human-cell ocular platform using iPSC-CECs for compound screening and therapy optimization. - Accelerate development of donor-independent corneal endothelial repair and next-generation neuroprotective therapies for ocular diseases. - Improve durability, consistency, and predictability of cell therapy performance by addressing key bottlenecks in cell resilience and functional stability, while enabling rapid expansion via the human-cell ocular platform.

6. References

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